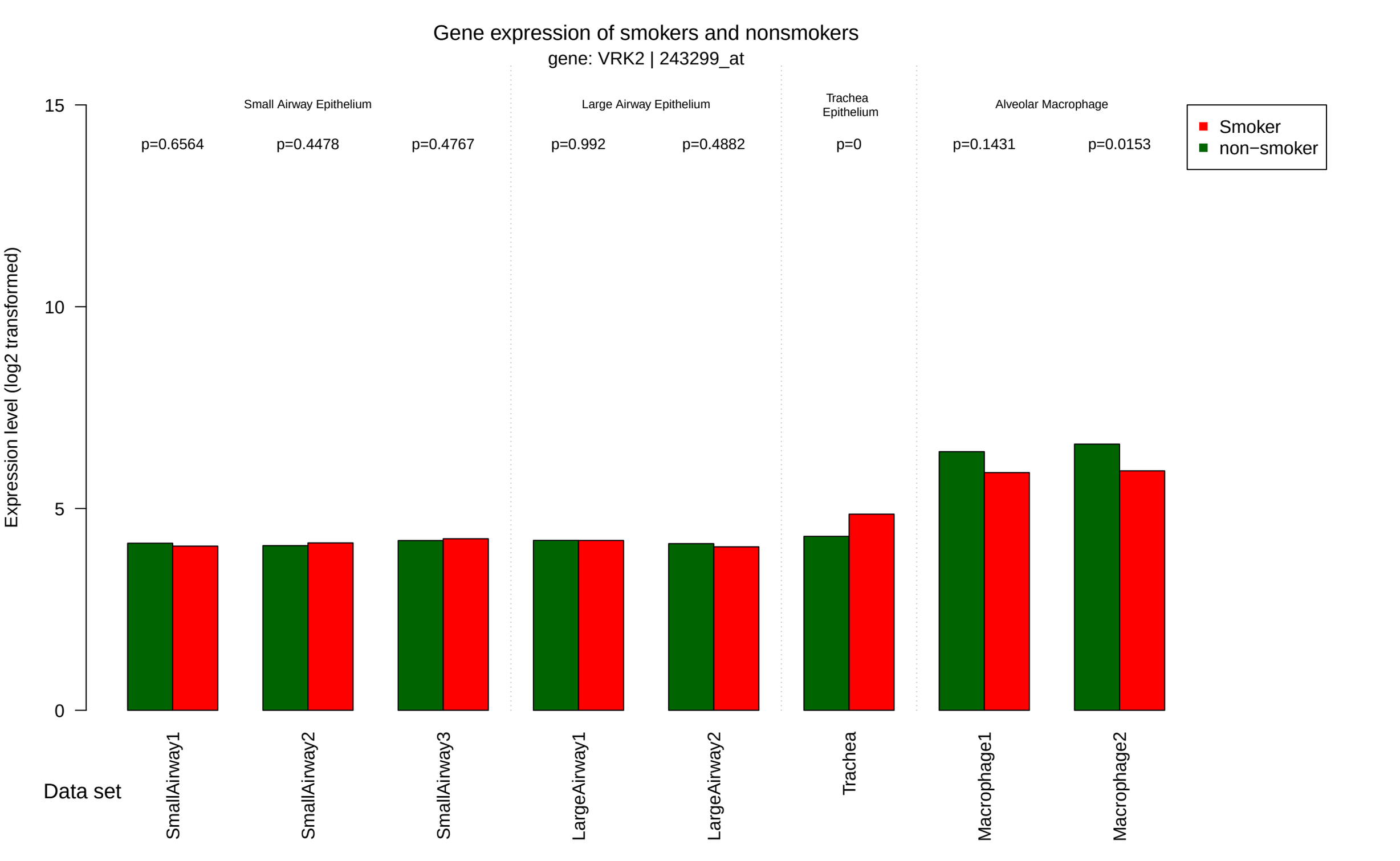
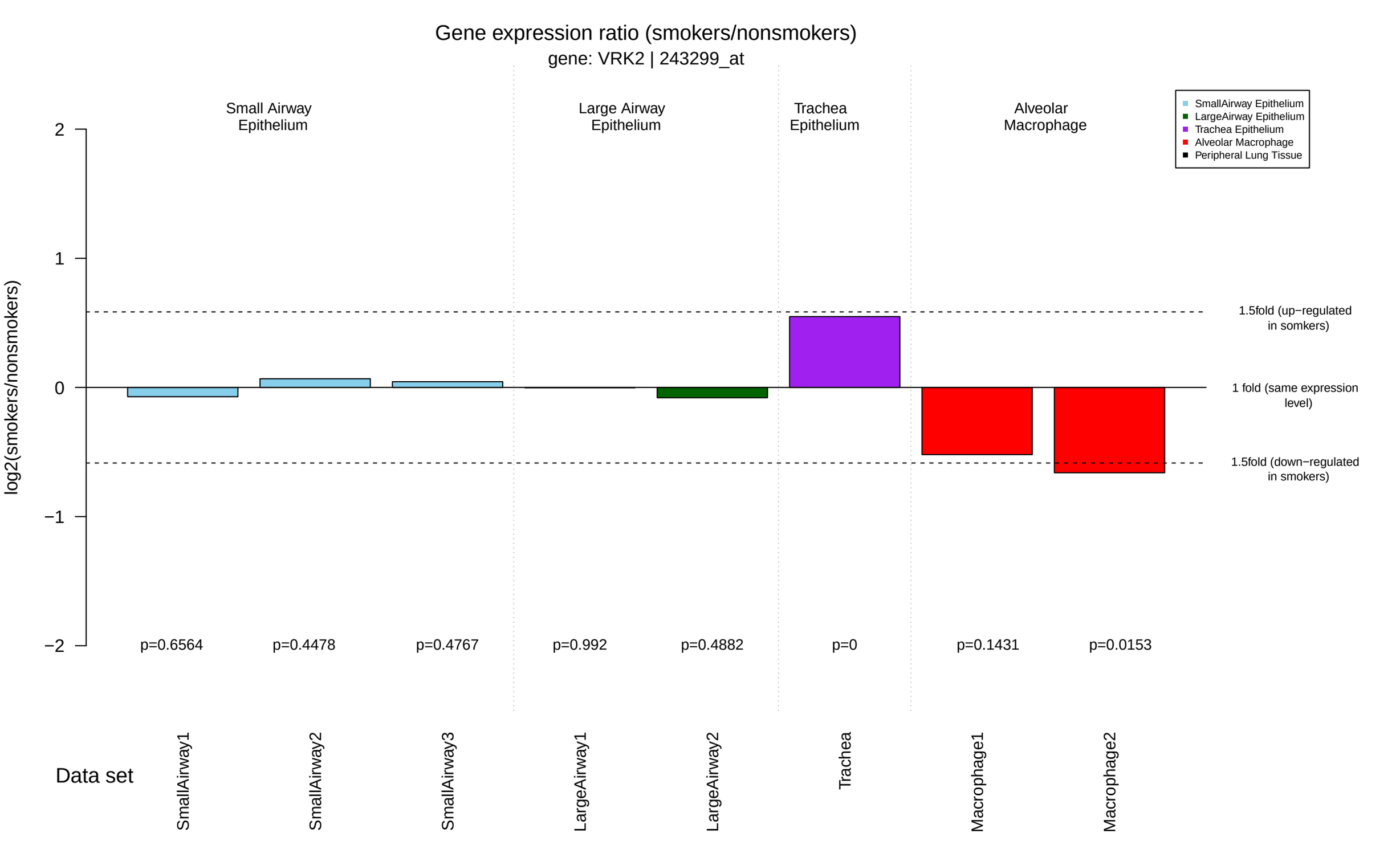
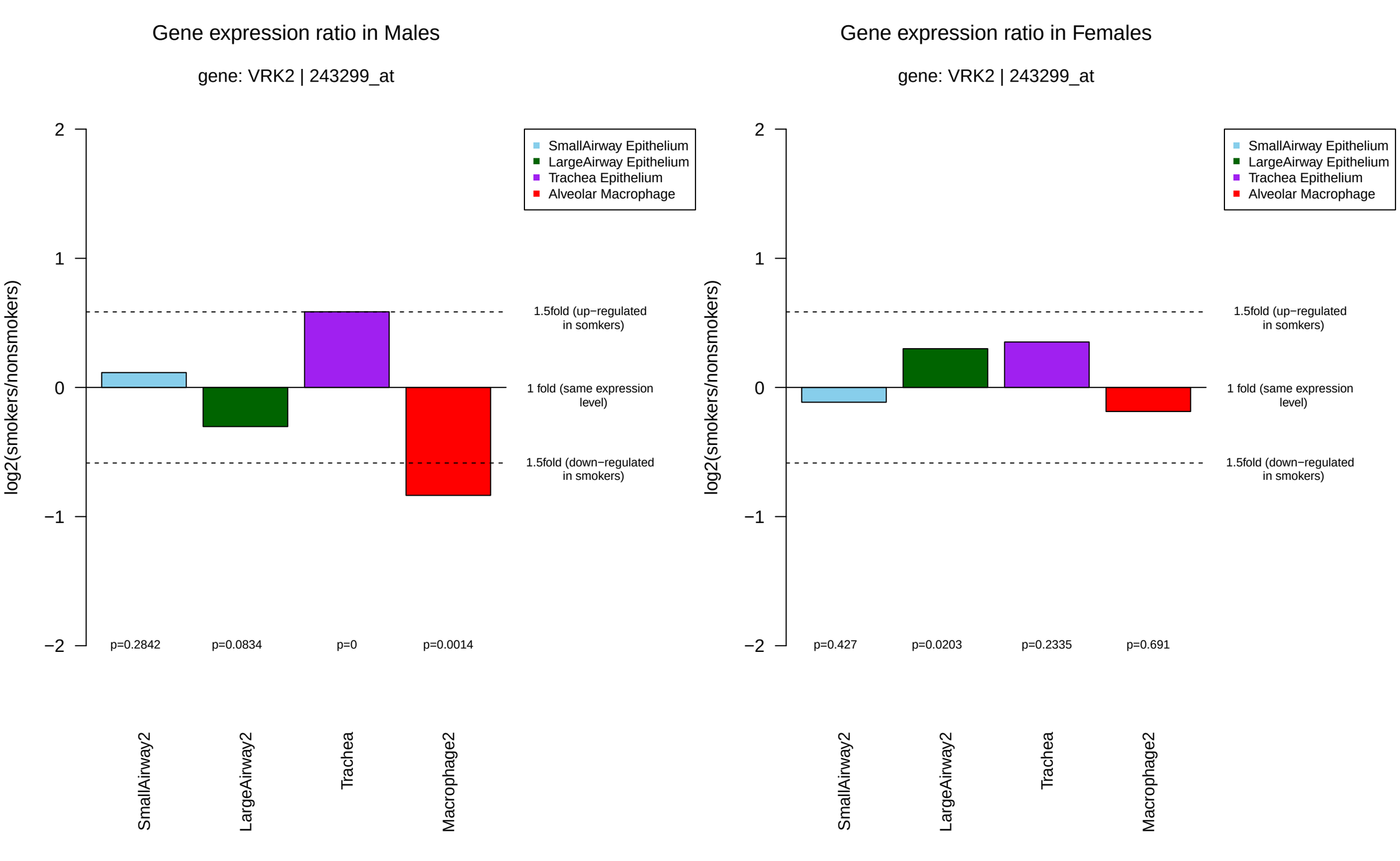
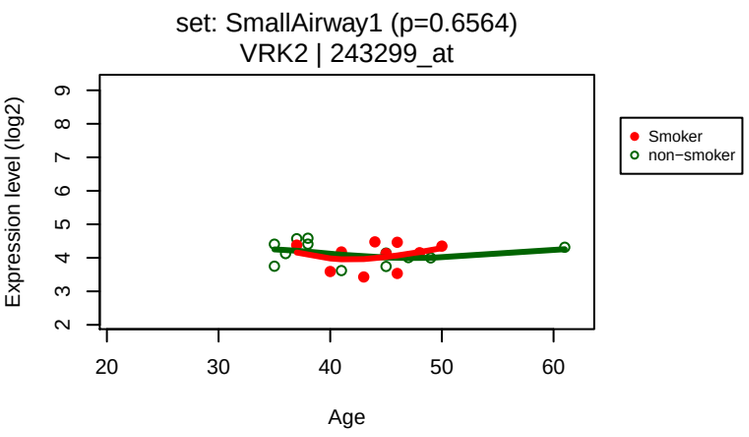
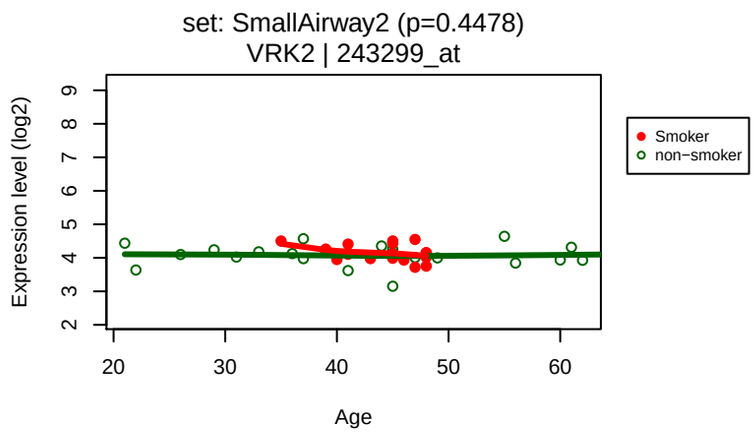
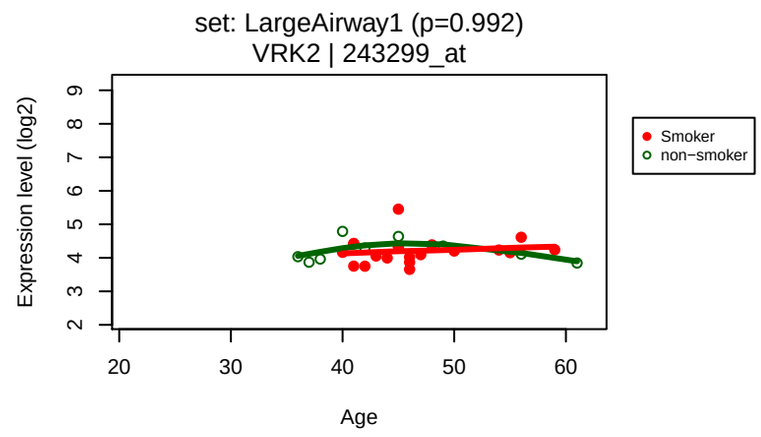

Gene expression of smokers and nonsmokers
gene: VRK2 | 243299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6564
p=0.4478
p=0.4767
p=0.992
p=0.4882
p=0
p=0.1431
p=0.0153
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

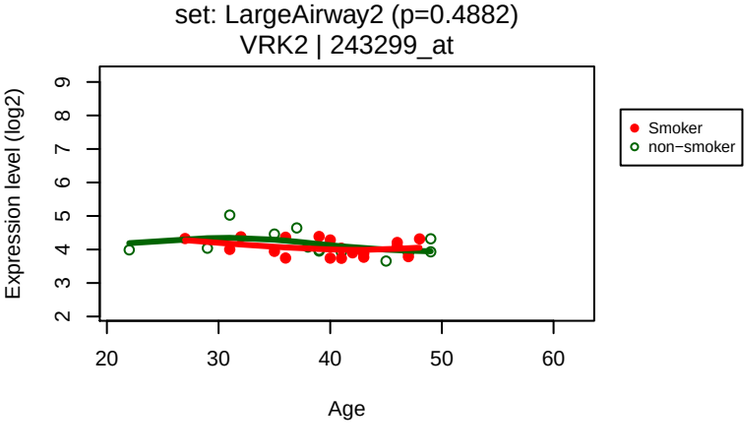

Gene expression ratio (smokers/nonsmokers)
gene: VRK2 | 243299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6564
p=0.4478
p=0.4767
p=0.992
p=0.4882
p=0
p=0.1431
p=0.0153
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

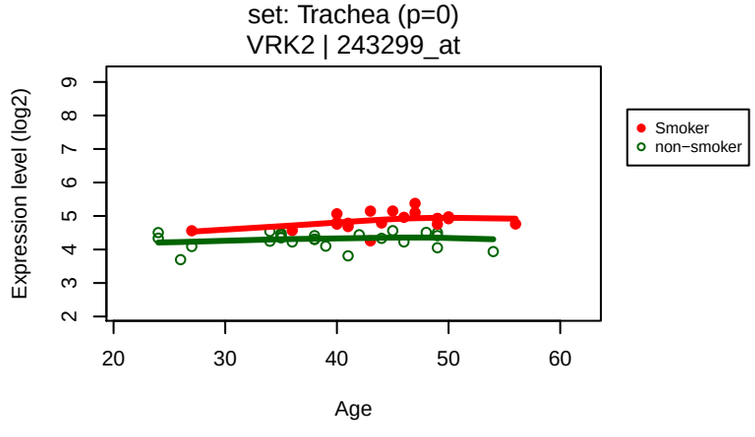

Gene expression ratio in Males
gene: VRK2 | 243299_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2842
p=0.0834
p=0
p=0.0014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VRK2 | 243299_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.427
p=0.0203
p=0.2335
p=0.691
SmallAirway2
LargeAirway2
Trachea
Macrophage2

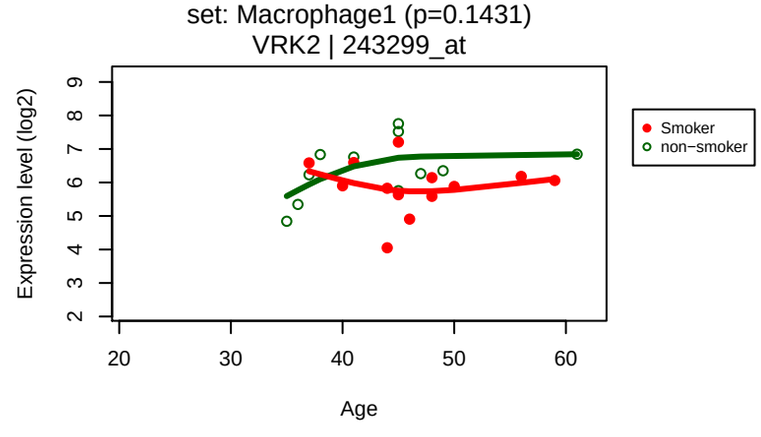

set: SmallAirway1 (p=0.6564)
VRK2 | 243299_at
Expression level (log2)
Age
Smoker
non−smoker

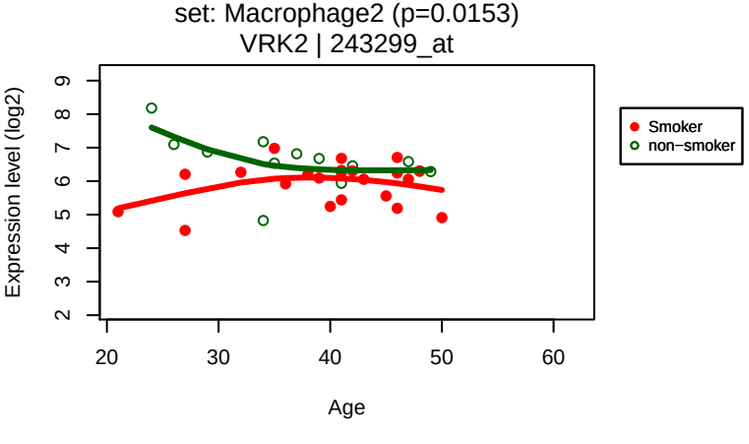

set: SmallAirway2 (p=0.4478)
VRK2 | 243299_at
Expression level (log2)
Age
Smoker
non−smoker

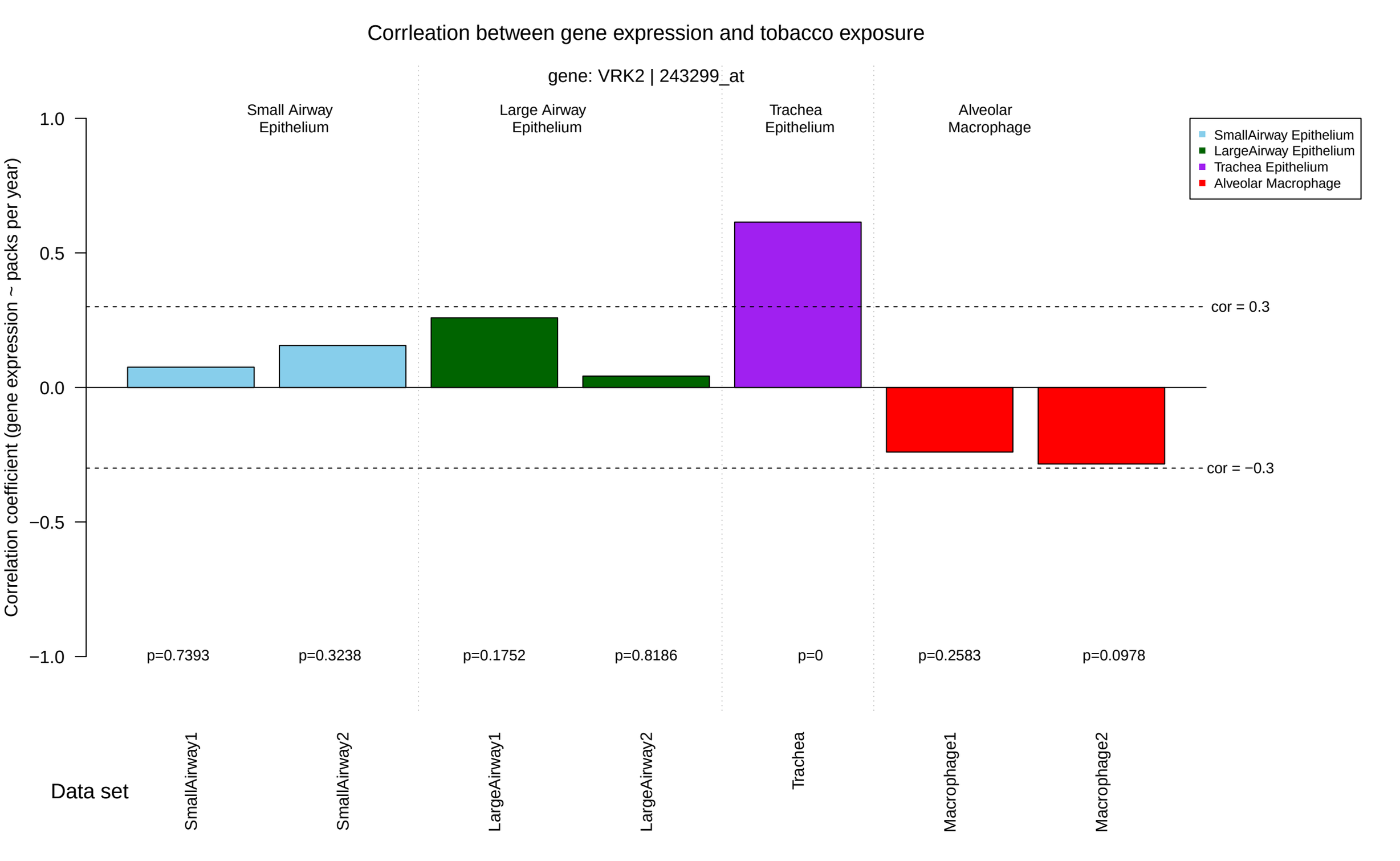

set: LargeAirway1 (p=0.992)
VRK2 | 243299_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4882)
VRK2 | 243299_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
VRK2 | 243299_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1431)
VRK2 | 243299_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0153)
VRK2 | 243299_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VRK2 | 243299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7393
p=0.3238
p=0.1752
p=0.8186
p=0
p=0.2583
p=0.0978
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2